Geometry of Nilpotent Orbits and Springer Theory

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1 Basics of Lie Theory

Let G be a connected reductive group over an algebraically closed field k. Recall the flag variety X is the moduli space of all Borel subgroups inside G. By choosing a Borel B, we can identify: $X \simeq G/B$.

Example. *For $G = \mathrm{GL}_n$, the flag variety is just the moduli space of all complete flags.*

Example. *For $G = \mathrm{Sp}_{2n} = \mathrm{Sp}(V, \omega)$, we have:*

$$X = \{0 \subset V_1 \subset \cdots \subset V_n = V_n^\perp \subset V_{n-1}^\perp \subset \cdots \subset V_1^\perp \subset V\}$$

*based on lectures by Zhiwei Yun

In fact there is a $\mathbb{P}^1$-family of choices of $\mathbb{P}^n$ after fixing the others. This is saying that we have a $\mathbb{P}^1$-fibration:

$$\mathbb{P}^1 \to X \to G/P_n$$

where P_n is the minimal parabolic corresponding to the special simple root α_n.

Example. *For $G = \mathrm{SO}_{2n+1} = \mathrm{SO}(V, q)$, the flag variety is:*

$$X = \{0 \subset V_1 \subset \cdots \subset V_n \subset V_n^\perp \subset V_{n-1}^\perp \subset \cdots \subset V_1^\perp \subset V\}$$

Example. *For $G = \mathrm{SO}_{2n} = \mathrm{SO}(V, q)$, naively we consider:*

$$Y = \{0 \subset V_1 \subset \cdots \subset V_n = V_n^\perp \subset V_{n-1}^\perp \subset \cdots \subset V_1^\perp \subset V\}$$

In this case Y has two components. In fact, consider $n = 1$ and consider the quadratic form xy, Y is simply two points. More generally, consider $\mathrm{Lag}(V)$ be the moduli space of Lagrangian subspaces, it has two components:

$$\mathrm{Lag}^+(V), \mathrm{Lag}^-(V)$$

It is easy to see that there are two isotropic subspaces between $V_{n-1}, V_{n-1}^\perp$, one lying in $\mathrm{Lag}^+(V)$ and the other line in $\mathrm{Lag}^-(V)$. Call these two V_n, V_n'. Note that we can then recover $V_{n-1} = V_n \cap V_n'$. This tells us the flag variety looks like:

$$X = \{V_1 \subset \cdots \subset V_{n-2} \subset V_n = V_n^\perp, V_n' = (V_n')^\perp \subset V_{n-2}^\perp \subset \cdots \subset V_1^\perp \mid \dim(V_n \cap V_n') = n-1\}$$

The V_n, V_n' correspond to the special vertices in the Dynkin diagram. In fact, we can also describe partial flag varieties in this way.

Suppose we take quadratic form:

$$q(x, y) = x_1 y_1 + \cdots + x_n y_n = 0$$

Let G acts on $\mathrm{Lag}^+(V)$. Let's choose the base point:

$$V_n = \{y_1 = y_2 = \cdots = y_n = 0\}$$

Then the stablizer of W equals:

$$P = \left\{ \begin{pmatrix} \mathrm{GL}(V_n) & * \\ & \mathrm{GL}(V_n) \end{pmatrix} \right\}$$

where the two $\mathrm{GL}(V_n)$ are contragradient to each other (i.e. $\mathrm{GL}(V_n)$ acts on $V_n^* \simeq V/V_n$). This is a maximal parabolic, with Levi quotient $\mathrm{GL}(V_n)$. On the level of Dynkin diagram, then embedding:

$$\mathcal{B}_{\mathrm{GL}(V_n)} \hookrightarrow X$$

corresponds to take the sub-diagram of the Dynkin diagram by discarding the vertex V_n.

Similarly, if we choose the vertex $V_1 \subset \mathbb{P}(V)$ consisting of $q(V_1, V_1) = 0$. This is a quadratic hypersurface in $\mathbb{P}(V)$, and the stablizer of V_1 looks like:

$$P = \left\{ \begin{pmatrix} * & * & * \\ & \mathrm{SO}(V_1^\perp/V_1) & * \\ & & * \end{pmatrix} \right\}$$

whose Levi quotient is $\mathrm{SO}(V_1^\perp/V_1)$.

Let T be a maximal tori, then let's denote the corresponding Weyl group to be $W(G,T) = N_G(T)/T$. These are not canonical. The canonical choice of torus is $T_B = B/U$, this is the *absolute Cartan* of G.

Example. *Let $G = \mathrm{GL}_n$. A choice of B is equivalent to a choice of a flag $V_\bullet$. Then T_B corresponds to the symmetry of the graded vector space $\mathrm{gr}_i V = V_i/V_{i-1}$.*

Now the correct definition of Weyl group is:

Definition 1.1. The abstract Weyl group $\mathbb{W}$ of G as a set is:

$$G \setminus (X \times X)$$

The length function is given by:

$$\ell(w) = \dim \mathcal{O}(w) - \dim X$$

This specifies the simple reflections in $\mathbb{W}$. Moreover, the identity element $\mathcal{O}(1) = \Delta X$.

To recover the group law, let's consider the convolution:

$$\mathcal{O}(w_1) * \mathcal{O}(w_2) = \{(x,x'') \in X \times X \mid \exists x' \in X, (x,x') \in \mathcal{O}(w_1), (x',x'') \in \mathcal{O}(w_2)\}$$

this is a union of orbits. The unique orbit with maximal dimension is defined to be $\mathcal{O}(w_1 * w_2)$, the unique orbit with minimal dimension is defined to be $\mathcal{O}(w_1 w_2)$.

Example. *Fix data $T \subset B$. A simple root α gives a minimal parabolic P_α. Let $X_\alpha = G/P_\alpha$, which sits inside $\mathbb{P}^1$-bundle: $X \to X_\alpha$. Then*

$$X \times_{X_\alpha} X = \Delta X \cup \mathcal{O}(s_\alpha)$$

Now let's see how $\mathbb{W}$ acts on T_{abs}. Given $w \in \mathbb{W}$, we choose $(B, B') \in \mathcal{O}(w)$ and we have correspondence:

$$T_{\mathrm{abs}} \simeq T_B \leftarrow B \cap B' \to T_{B'} \simeq T_{\mathrm{abs}}$$

then $\Gamma(w) = \mathrm{Im}(B \cap B' \to T_{\mathrm{abs}} \times T_{\mathrm{abs}})$.

With this action, we can define the *abstract root system* (consisting the data of roots as a collection of simple roots). Let's again pick $T \subset B$, and consider composition:

$$\iota_{T,B} : T \hookrightarrow B \twoheadrightarrow T_{\mathrm{abs}}$$

We can then transport the data $\Delta(G,B,T) \subset \Phi(G,T) \subset X^*(T)$ to $\Delta_{\mathrm{abs}} \subset \Phi_{\mathrm{abs}} \subset X^*(T_{\mathrm{abs}})$.

2 Jacobson-Morozov Theorem

Let $\mathcal{N} \subset \mathfrak{g}$ be the nilpotent cone. The goal is to prove:

Theorem 2.1. *(Jacobson-Morozov) THe map:*

$$\{\mathfrak{sl}_2\text{-triples}\}/G \to \mathcal{N}/G$$

$$(e, f, h) \mapsto e$$

is a bijection.

Let's start by showing existence. The idea is to reduce to the case $\mathfrak{gl}_n$. For type A, let E be a nilpotent matrix, we can put E in Jordan normal form and it is not hard to pick H and F. For the general case, we first prove a weaker version:

Lemma 2.2. *There exists f', h' such that $[h', e] = 2e, [e, f'] = h'$.*

Proof. Choose a faithful representation $G \to \mathrm{GL}_n$, which gives a faithful representation $\mathfrak{g} \to \mathfrak{gl}_n$. The trace pairing on $\mathfrak{gl}_n$ restricts to a nondegenerate equivariant (under the group action) pairing on $\mathfrak{g}$. It follows that:

$$\mathfrak{gl}_n = \mathfrak{g} \oplus \mathfrak{g}^\perp$$

Since $\mathfrak{g}^\perp$ is G-equivariant, so $[\mathfrak{g}, \mathfrak{g}^\perp] \subset \mathfrak{g}^\perp$. Now let E be the image of $e \in \mathfrak{g}$ in $\mathfrak{gl}_n$, we can accordingly pick H, F satisfying the $\mathfrak{sl}_2$-triple condition. Now let's write:

$$F = f' + f'', H = h' + h''$$

for $f', h' \in \mathfrak{g}$, $f'', h'' \in \mathfrak{g}^\perp$. It follows that:

$$2e = [H, E] = [h' + h'', e] = [h', e]$$

$$[e, f' + f''] = [E, F] = H = h' + h''$$

Look at $\mathfrak{g}$-component: $[e, f'] = h'$. However, we may not have:

$$[h', f'] = -2f'$$

□

To enforce the last equality, we will set: $h = h'$ and perturb f' to $f = f' + v$. Note that then we need $[e, v] = 0$ and $[h, f' + v] = -2(f' + v)$, in other words:

$$(\mathrm{ad}_h + 2)v = -[h, f'] - 2f'$$

Notice that $\mathrm{ad}_h + 2$ acts on the space $\mathfrak{g}^e$, and $-[h, f'] - 2f' \in \mathfrak{g}^e$. Therefore it suffices to show that $\mathrm{ad}_h + 2$ is invertible on $\mathfrak{g}^e$.

So we need to organize $\mathfrak{g}^e$. Indeed, let's consider the descending filtration:

$$F^n\mathfrak{g}^e \coloneqq \mathfrak{g}^e \cap \mathrm{im}\ (\mathrm{ad}_e)^n$$

which is compatible with the ad$h$-action. In fact, from the story for $\mathfrak{sl}_n$, we expect:

Lemma 2.3. *Let $0 \neq v = (\mathrm{ad}_e)^k x$, such that $v \in \mathfrak{g}^e \cap \mathrm{im}\ (\mathrm{ad}_e)^k$. Then $(\mathrm{ad}_h)v \in kv + \mathrm{im}\ (\mathrm{ad}_e)^{k+1}$. In other words, ad_h acts on $\mathrm{gr}^k_F \mathfrak{g}^e$ by k.*

Proof. We will use the relation: $[\mathrm{ad}_h, \mathrm{ad}_e] = 2\mathrm{ad}_e, [\mathrm{ad}_e, \mathrm{ad}_{f'}] = \mathrm{ad}_h, (\mathrm{ad}_e)^{k+1}x = 0$. So we can compute:

$$\begin{aligned}(\mathrm{ad}_e)^k(\mathrm{ad}_h)x &\in -(\mathrm{ad}_e)^k(\mathrm{ad}_{f'})(\mathrm{ad}_e)x + \mathrm{im}\ (\mathrm{ad}_e)^{k+1} \\ &= -\sum_{i=1}^{k}(\mathrm{ad}_e)^{k-i}(\mathrm{ad}_h)(\mathrm{ad}_e)^i x + \mathrm{im}\ (\mathrm{ad}_e)^{k+1} \\ &= -\sum_{i=1}^{k}((\mathrm{ad}_e)^k(\mathrm{ad}_h)x + 2i(\mathrm{ad}_e)^k x) + \mathrm{im}\ (\mathrm{ad}_e)^{k+1} \\ &= -k(\mathrm{ad}_e)^k(\mathrm{ad}_h)x - k(k+1)(\mathrm{ad}_e)^k x + \mathrm{im}\ (\mathrm{ad}_e)^{k+1}\end{aligned}$$

Therefore $(\mathrm{ad}_e)^k(\mathrm{ad}_h)x \in k(\mathrm{ad}_e)^n x + \mathrm{im}\ (\mathrm{ad}_e)^{k+1}$ and $\mathrm{ad}_h(v) \in kv + \mathrm{im}\ (\mathrm{ad}_e)^{k+1}$ as desired. □

As a consequence, we see that ad_h has non-negative eigenvalues, and ad_h+2 acts invertibly on $\mathfrak{g}^e$.

The next goal is to prove uniqueness. First of all, the above proof shows that f is uniquely determined by the choice of e, h: suppose we have two choices f, f', then $f - f' \in \mathfrak{g}^e$. But we know:

$$(\mathrm{ad}_h + 2)(f - f') = 0 \implies f - f' = 0$$

since eigenvalues ≥ 2. So it suffices to show that for any two choices h, h', there exists an element in $Z_G(e)$ taking one to the other.

Observe that the filtration on $\mathfrak{g}^e$ by $\mathrm{im}\ (\mathrm{ad}e)^n$ admits a splitting, and $\mathfrak{g}^e$ is a direct sum where $\mathrm{ad}h$ acts on the n-th summand by n. Now let:

$$\mathfrak{u} = \mathfrak{g}^e \cap \mathrm{im}\ (\mathrm{ad}e)$$

This is a subalgebra of $\mathfrak{g}^e$ and is nilpotent (since it has positive h-weights). Now Engel's theorem ensures that it can be brought into strictly upper-triangular form. Now let $U = \exp(\mathfrak{u}) \subset C_G(e)$ be the connected closed Lie subgroup. It suffices to show that U acts transitively on the set of h' such that it extends e to a $\mathfrak{sl}_2$-triple.

Lemma 2.4. *The affine space $h+\mathfrak{u}$ is precisely the set of all h' which extends e to a $\mathfrak{sl}_2$-triple.*

Proof. If (e, f, h) and (e, h', f') are $\mathfrak{sl}_2$-triples, then clearly $h - h' \in \mathfrak{g}^e$ and $h - h' = [e, f - f']$, so that $h - h' \in \mathfrak{u}$. Conversely, suppose $h' = h + [e, x]$, we take the first approximation to f' to be $f + x$, so that:

$$[e, f + x] = h'$$

and then we can adjust it according to the existence part. □

Lemma 2.5. *The action of U on $h + \mathfrak{u}$ is simply transitive.*

Proof. Consider the action map:

$$U \to h + \mathfrak{u}$$

$$g \mapsto \mathrm{Ad}(g)h$$

The derivation of this map is clearly an isomorphism. To see the original map is an isomorphism requires to filter U and $\mathfrak{u}$ by the heights of roots, but what happens at the level of associated graded coincides with the linear version. We omit the full detail here. □

This concludes the proof of uniqueness.

Remark. *1. In the weight decomposition $\mathfrak{g} = \bigoplus \mathfrak{g}_i$ where $\mathrm{ad}h$ acts on $\mathfrak{g}_i$ by i. Then the subspace $\mathfrak{g}_{\geq i}$ only depends on e (and not on the choice of h). In particular, given a nilpotent e, we can consider a parabolic $\mathfrak{g}_{\geq 0}$ and a corresponding parabolic subgroup $P \subset G$, which only depends on e. As a consequence, $C_G(e)$ acts on P, and since P is self-normalizing, we have $C_G(e) \subset P$. This is obvious on the Lie algebra level, but $C_G(e)$ is not necessarily connected.*

2. The subalgebra $\mathfrak{g}_0$ is reductive, and denote by $G_0 \subset G$ corresponding connected reductive group. Then:

$$C_G(e) = U \rtimes C_{G_0}(e)$$

To see this , we consider the $C_G(e)$ action on $h + \mathfrak{u}$. This is a transitive action, we the stablizer of h equals $C_{G_0}(e)$. The result then follows by orbit stablizer. Moreover, we note that $C_{G_0}(e)$ is reductive, and the above is the Levi-decomposition for the group $C_G(e)$.

3 Bala-Carter Classification

The goal of the theory is to find a parametrization of the nilpotent orbits for a reductive Lie algebra $\mathfrak{g}$. Recall we have:

$$\{\mathbb{Z}\text{-grading of } \mathfrak{g}\}/G \xleftarrow{a} \{\mathfrak{sl}_2\text{-triples in } \mathfrak{g}\}/G \xrightarrow{b} \mathcal{N}/G$$

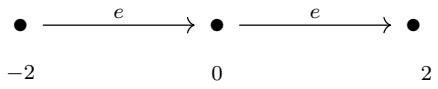
We already saw that b is a bijection. The starting point of the theory is that:

Theorem 3.1 (Malcev)**.** *The map a is injective.*

Proof. It suffices to show that given any two pairs $(e, h), (e', h)$ extendable to a $\mathfrak{sl}_2$-triple, then we necessarily have they're are conjugate. In particular, note that we can only conjugate by elements in $C_G(h)$. In fact, we will prove that we will show that e, e' lie in the same G_0-orbit, where G_0 is the connected reductive group corresponding to $\mathfrak{g}_0$. For this, we only need to prove that $G_0 \cdot e \subset \mathfrak{g}_2$ is Zariski dense (therefore $G_0 \cdot e \cap G_0 \cdot e' \neq \emptyset$). Note that the action map $G_0 \to \mathfrak{g}_2$ as tangent map $\mathfrak{g}_0 \to \mathfrak{g}_2$ given by $X \mapsto [X, e] = -\mathrm{ad}_e(X)$, which is surjective. We know that $G_0 \to \mathfrak{g}_2$ is dominant, and therefore has dense image. □

The next goal is to identify the image of a.

Example. *Let $\mathfrak{g} = \mathfrak{sl}_3$. It has three nilpotent orbits described by Jordan types. The regular orbit can be described as the grading on the 3-dimensional standard representation V equipped with the nilpotent the nilpotent endormorphism:*



whose corresponding $\mathbb{Z}$-grading on $\mathfrak{g}$ is given by:

$$\begin{pmatrix} 0 & 2 & 4 \\ -2 & 0 & 2 \\ -4 & -2 & 0 \end{pmatrix}$$

For the subregular (or equivalently the minimal) orbit, then the $\mathbb{Z}$-grading on $\mathfrak{g}$ is given by:

$$\begin{pmatrix} 0 & 2 & 1 \\ -2 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}$$

Up to scalar, this is simply the rearrangment of the above matrix. However, it's clear that there are many gradings that do not come from a $\mathfrak{sl}_2$-triple.

The idea of the Bala-Carter classification is the following. For a nilpotent element e, we extend it into a $\mathfrak{sl}_2$-triple $\{e, f, h\}$, then we seek to find a minimal Levi subalgebra $\mathfrak{l}$ (of some parabolic subalgebra) containing this $\mathfrak{sl}_2$-triple. Then the $\mathbb{Z}$-grading on $\mathfrak{l}$ coming from the h-action will be "distinguished" in a way that we will explain below. But first let's explain how to find the minimal Levi $\mathfrak{l}$.

Given a $\mathfrak{sl}_2$-triple $\{e, f, h\}$ in $\mathfrak{g}$, we consider $Z_{\mathfrak{g}}(e, f, h)$. This will be a reductive subalgebra. Let's pick $\mathfrak{s} \subset Z_{\mathfrak{g}}(e, f, h)$ a maximal toral subalgebra, we set $\mathfrak{l} = Z_{\mathfrak{g}}(\mathfrak{s})$.

Now let's explain how to extend a $\mathbb{Z}$-grading on $\mathfrak{l}$ to a $\mathbb{Z}$-grading on $\mathfrak{g}$. Given a $\mathfrak{sl}_2$-triple in $\mathfrak{g}$, we have an associated Levi $\mathfrak{l}$. Start with a $\mathbb{Z}$-grading on $\mathfrak{l}$ (which comes from a $\mathfrak{sl}_2$-triple), we can construct a $\mathbb{Z}$-grading on $\mathfrak{g}$ as follows. Note the data of a $\mathbb{Z}$-grading on $\mathfrak{l}$ comes from the action of $\mathrm{ad}(h)$, so that this actually comes from a $\mathbb{G}_m$-action that factors through L_{ad}. So I have a map $\mathbb{G}_m \to L_{\mathrm{ad}}$. Moreover, since the semisimple element h sits inside a $\mathfrak{sl}_2$-triple, we acutally have $\mathbb{G}_m \to [L, L] \subset G$. This action gives a $\mathbb{G}_m$-action on $\mathfrak{g}$.

Definition 3.2. A *distinguished $\mathbb{Z}$-grading* on $\mathfrak{g}$ is a decomposition $\mathfrak{g} = \bigoplus_{n \in \mathbb{Z}} \mathfrak{g}_n$ satisfying:

1. $\mathfrak{g}_{\mathrm{odd}} = 0$
2. $\dim \mathfrak{g}_0 - \dim \mathfrak{z}(\mathfrak{g}) = \dim \mathfrak{g}_2$

Example. *Consider $\mathfrak{g} = \mathfrak{sl}_n$. For the principal $\mathfrak{sl}_2$-triple $\{e, f, h\}$, the Levi $\mathfrak{l} = \mathfrak{g}$. Now the action of $\mathrm{ad}(h)$ on $\mathfrak{g}$ gives a distinguished $\mathbb{Z}$-grading, with $\dim \mathfrak{g}_0 = \dim \mathfrak{g}_2 = n - 1$. In fact, this is the only distinguished $\mathbb{Z}$-grading on $\mathfrak{g}$.*

The main result is the following:

Theorem 3.3 (Bala-Carter)**.** *There is a bijection:*

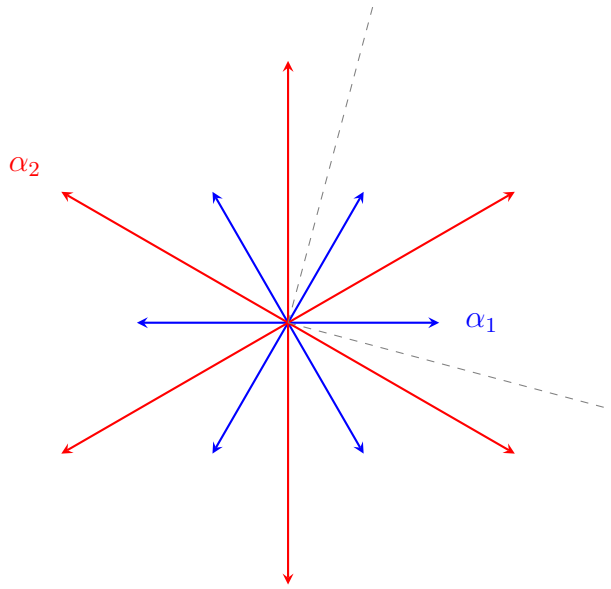
$$\{(\mathfrak{l}, \text{distinguished } \mathbb{Z}\text{-grading on } \mathfrak{l})\}/G \xrightarrow{\sim} \{\mathbb{Z}\text{-grading comes from a } \mathfrak{sl}_2\text{-triple}\}/G$$

In particular, both sides biject with the set of nilpotent orbits of $\mathfrak{g}$*.*

Example. *The following table gives the number of distinguished* $\mathbb{Z}$*-gradings for exceptional groups:*

G	number of distinguished $\mathbb{Z}$-gradings modulo conjugacy
G_2	2
F_4	4
E_6	3
E_7	6
E_8	11

Example. *Let's describe the distinguished* $\mathbb{Z}$*-grading on* G_2 *called* $G_2(a_1)$*.*

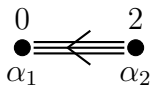


Consider the $\mathbb{Z}$*-grading obtained from projecting to the one-dimensional space containing the vertical long root. Then:*

$$\dim \mathfrak{g}_0 = \dim \mathfrak{g}_2 = \dim \mathfrak{g}_{-2} = 4$$

$$\dim \mathfrak{g}_4 = \dim \mathfrak{g}_{-4} = 1$$

Equivalently, we can draw the Dynkin diagram of G_2 *and give each vertex a number (which is the grading of the corresponding root space, zero omitted).*



It is clear that $\mathfrak{g}_0 \simeq \mathfrak{gl}_2$ *and* $G_0 \simeq \mathrm{GL}_2$*. Then as a* G_0*-representation, we know:*

$$\mathfrak{g}_2 \simeq \mathrm{Sym}^3 \otimes \det^{-1}$$

and the marked vertex is the lowest weight vector. It is also straightforward to see $\mathfrak{g}_4 \simeq \det$*.*

Now let's describe $e \in \mathfrak{g}_2$*. By the proof of Malcev's theorem, we know this up to conjugation is nothing but the open dense orbit of* G_0 *acting on* $\mathfrak{g}_2$*. By viewing* $\mathfrak{g}_2 \simeq \mathrm{Sym}^3 \otimes \det^{-1}$*, which is isomorphic to binary cubic forms. Therefore, the action of* G_0 *on* $\mathbb{P}(\mathfrak{g}_2) \simeq \mathbb{P}(\mathrm{Sym}^3 V) \simeq (\mathbb{P}^1)^3/S_3$ *comes from the* G_0*-action on* $(\mathbb{P}^1)^3$*. Now that we can view the generic* $e \in \mathbb{P}(\mathfrak{g}_2)$*, and then we have a natural map:*

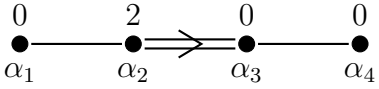
$$\mathrm{stab}_{G_0}(e) \to S_3$$

by considering the action. It turns out that this is actually an isomorphism. This gives:

$$C_G(e) = (\text{unipotent group}) \rtimes S_3$$

and therefore: $\pi_0(C_G(e)) \simeq S_3$*. This is the smallest non-abelian finite group arising in this way.*

Example. *There is a distinguished* $\mathbb{Z}$*-grading on* F_4 *called* $F_4(a_3)$*, according to the following weighted Dynkin diagram.*



We can still decompose:

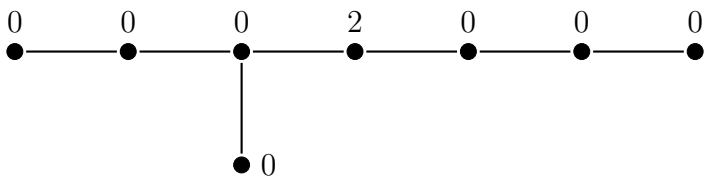
$$\mathfrak{g} = \bigoplus_{n \in \mathbb{Z}} \mathfrak{g}_{2n}$$

We have $G_0 \sim \mathrm{GL}_2 \times \mathrm{GL}_3$ *(this means up to center and central isogeny), and* $\mathfrak{g}_2 \simeq U_2 \otimes \mathrm{Sym}^2 V_3$ *for* U_2, V_3 *standard representations. Let's descirbe the corresponding nilpotent element* e*, or equivalently, the open dense orbit of* G_0 *on* $\mathfrak{g}_2$*. As before, we identify* $\mathbb{P}(\mathfrak{g}_2)$ *as the moduli space of one-dimensional pencil of quadratic curves in* $\mathbb{P}^2$*. Then the genericity of* e *translates to the requirement that the corresponding pencil has its base points consisting of four distinct points. We thus get a map:*

$$\mathrm{Stab}_{G_0}(e) \to S_4$$

which is in fact an isomorphism.

Example. *There is a distinguished* $\mathbb{Z}$*-grading on* E_8 *called* $E_8(a_7)$*:*



In this case $G_0 \sim \mathrm{GL}_5 \times \mathrm{GL}_4$ *and* $\mathfrak{g}_2 \simeq \Lambda^2 V_5 \otimes U_4$*. By a similar mannaer we will get an isomorphism:* $\mathrm{Stab}_{G_0}(e) \simeq S_5$*.*

4 Steinberg's Formula

For a simple Lie algebra $\mathfrak{g}$, we have a partially ordering on the sets on nilpotent orbits:

$$\{0\} < \mathcal{O}_{\min} < \cdots < \mathcal{O}_{\text{subreg}} < \mathcal{O}_{\text{reg}}$$

We have:

$$\dim(\mathcal{O}_{\text{reg}}) = \dim(\mathcal{N}) = \dim(G/T)$$
$$\dim(\mathcal{O}_{\text{subreg}}) = \dim(\mathcal{O}_{\text{reg}}) - 2$$
$$\dim(\mathcal{O}_{\min}) = 2(h^\vee - 1)$$

for $h^\vee$ the dual Coxeter number. Note that each nilpotent orbit is symplectic, hence has even dimension.

Theorem 4.1. *(Steinberg's dimension formula) For each nilpotent orbit $\mathcal{O} \subset \mathcal{N}$, we have:*

$$\operatorname{codim}_{\mathcal{N}}(\mathcal{O}) = 2\dim(\mathcal{B}_e)$$

To prove this, let's consider the Steinberg variety:

$$\mathsf{St} = \widetilde{\mathcal{N}} \times_{\mathcal{N}} \widetilde{\mathcal{N}} = \coprod_{w \in W} \mathsf{St}(w)$$

where $\mathsf{St}(w) = \overline{T^*_{\mathcal{O}(w)}(\mathcal{B} \times \mathcal{B})}$ is a Lagrangian in $T^*(\mathcal{B} \times \mathcal{B})$. It is clear that $\dim \mathsf{St} = \dim \mathcal{N}$. However, there is another presentation of St:

$$\mathsf{St} = \{(e, B, B') \mid e \in \mathfrak{n}_B \cap \mathfrak{n}_{B'}\} = \coprod_{\mathcal{O}_e \in \mathcal{N}/G} \mathsf{St}_e$$

for St_e sits inside fibre bundle:

$$\mathcal{B}_e \times \mathcal{B}_e \to \mathsf{St}_e \to \mathcal{O}_e$$

These two perspectives give me parametrizations:

$$\operatorname{Irr}(\mathsf{St}) \longleftrightarrow W$$

and

$$\operatorname{Irr}(\mathsf{St}) \longleftrightarrow \coprod_{\mathcal{O}_e \in \mathcal{N}/G} (\operatorname{Irr}_{\text{top}}(\mathcal{B}_e) \times \operatorname{Irr}_{\text{top}}(\mathcal{B}_e))/C_G(e)$$

Of course we will prove Springer fibre is pure dimension and thus we will drop the subscript top in the future.

As a consequence, we get a map:

$$\Phi : W \to \mathcal{N}/G$$

Given $w \in W$, we pick any $(B, B') \in \mathcal{O}(w)$, and this determines a space $\mathfrak{n}_B \cap \mathfrak{n}_{B'}$. The image of w will then be the nilpotent orbit $\mathcal{O}_e$ who is dense in $\mathfrak{n}_B \cap \mathfrak{n}_{B'}$. The fibres of Φ partitions W, and are called S-cells. Let's describe this map in type A: combinatorially, this is a map from an element in S_n to a partition of n.

1. Consider $1 \in S_n$, then the intersection will be a maximal unipotent and thus $\Phi(1) = \mathcal{O}_{\text{reg}} = (n)$.

2. Consider $w_0 \in S_n$, then the intersection will be trivial and $\Phi(w_0) = (1^n)$.

3. The general recipe is the following. Given $\sigma \in S_n$, let's consider:

$$\begin{pmatrix} 1 & 2 & \cdots & n \\ \sigma(1) & \sigma(2) & \cdots & \sigma(n) \end{pmatrix}$$

   Then let's consider $S_1, \cdots, S_k$ be ordered increasing sequences partition $\{\sigma(1), \cdots, \sigma(n)\}$, such that $\sum_{i=1}^k |S_i|$ is maximal among all choices. Then the output partition will be the partition corresponding to $S_1, \cdots, S_k$. For example, consider $\sigma = (13425)$, it will output partition $\lambda = (2, 1, 1, 1)$.

In type A, the more refined map is precisely the *Robinson-Schensted correspondence*:

$$S_n \longleftrightarrow \text{SYT}(\lambda) \times \text{SYT}(\lambda)$$

This almost proves the Steinberg's formula, from the fibre bundle we consider. However, there is only one gap: we do not know that St_e is nonempty in priori. In other words, we must prove that for every e, there exists B, B' such that $\mathcal{O}_e$ is dense in $\mathfrak{n}_B \cap \mathfrak{n}_{B'} \iff \Phi$ is surjective.

Observation: Let L be a Levi quotient associated to some parabolic $P \subset G$ and assume Φ_L is surjective. By fixing a section $L \to P$ we may write $\mathcal{N}_L \subset \mathcal{N}$ and $W_L \subset W$. Now for any orbit $\mathcal{O}_e^L$ in L, we can find $w \in W_L$ we can find $\Phi_L(w) = \mathcal{O}_e^L$. One might expect to hope that $\Phi(w) = \mathcal{O}_e$. But this usually fails: for example, if $e = 0$ and $P = B$, then $L = T$ and thus $\mathcal{O}_e^T = 0 = \Phi_T(1)$ but $0 \neq \Phi(1)$. But it is true that $0 = \Phi(w_0)$ for the longest element w_0. This hints how we should proceed.

So the pattern is the following. All nilpotents coming from a smaller Levi should also come from the Weyl groups of that Levi. The remaining orbits will be distinguished. But the distinguished orbits will then come from longest element of the Weyl groups of the Levi. Let's now formalize these ideas.

Lemma 4.2. *Assume Φ_L is surjective, then $\mathcal{N}_L \subset \text{im } \Phi$.*

Proof. Let $e \in \mathcal{N}_L$ and assume $\mathcal{O}_e^L = \Phi_L(w)$ for $w \in W_L \subset W$. Therefore there exists $\overline{B}, \overline{B}' \subset L$ such that $\mathcal{O}_e^L$ is dense in $\mathfrak{n}_B \cap \mathfrak{n}_{B'}$. Choose P' the *opposite parabolic* of P so that $P' \cap P = L$. Let B, B' be preimages of $\overline{B}, \overline{B}'$ under $P \to L$ and $P' \to L$ respectively. This tells us that:

$$\mathfrak{n}_B \cap \mathfrak{n}_{B'} = \mathfrak{n}_{\overline{B}} \cap \mathfrak{n}_{\overline{B}'}$$

because $P \cap P' = L$. Therefore if $(B, B') \in \mathcal{O}(w')$, then $\Phi(w') = \mathcal{O}_e$. □

Definition 4.3. Let $P \subset G$ be a parabolic. We call the dense nilpotent orbit inside $\mathfrak{n}_P$ the *Richardson class* of P.

Lemma 4.4. *The Richardson class is $\Phi(w_{P,0})$ for $w_{P,0}$ the longest element of $W_P \subset W$.*

Proof. Choose $B \subset P$ with $\overline{B} \subset L$ its image in the Levi. Let $\overline{B}'$ the opposite Borel to $\overline{B}$ in L, and B' the preimage of $\overline{B}'$ in P. Then $(B, B') \in \mathcal{O}(w_{P,0})$ and $\mathfrak{n}_B \cap \mathfrak{n}_{B'} = \mathfrak{n}_P$. Since Richardson class is dense in $\mathfrak{n}_P$, we know it must equal $\Phi(w_{P,0})$. □

Proof. (of surjectivity of Φ). We use Bala-Carter classification. Suppose $\mathcal{O}$ is not distinguished, then we know it must be induced from its corresponding Levi and we're done. Otherwise $\mathcal{O}_e$ is distinguished, we extend it into a $\mathfrak{sl}_2$-triple $\{e, f, h\}$ and obtain a grading: $\mathfrak{g} = \bigoplus_{n\in\mathbb{Z}} \mathfrak{g}_{2n}$ with $e \in \mathfrak{g}_2$. We show that (during the proof of Jacobson-Morozov) $\mathcal{O}_e \cap \mathfrak{g}_{\geq 2} \subset \mathfrak{g}_{\geq 2}$ is dense; it equals $G_0 \cdot e + \mathfrak{g}_{\geq 4}$ and $G_0 \cdot e$ is dense in $\mathfrak{g}_2$. In particular $\mathcal{O}_e$ is the Richardson class corresponding to parabolic whose Lie algebra if $\mathfrak{g}_{\geq 0}$. We're therefore done. □

5 Borel-Moore Homology

Let's briefly recall the setting of Borel-Moore homology. Let X be a nice space, we can then form a chain complex of locally finite linear combinations of simplices. The corresponding homology group will be called Borel-Moore homology groups. It will have the properties:

1. If X is compact, then $H_*^{\mathrm{BM}}(X) = H_*(X)$.

2. There is a cap product:

$$\cap : H_p^{\mathrm{BM}}(X) \times H^q(X) \to H_{p-q}^{\mathrm{BM}}(X)$$

   Geometrically, a Borel-Moore class can be represented by a degree p distribution on de Rham complex (i.e. a linear functional $\Omega_c^p(X) \to \mathbb{R}$). Let's say the distribution is given by integrating along a differential form μ of degree $\dim X - p$, then the cap product is given by:

$$(\mu, \omega) \mapsto \mu \wedge \omega \in \Omega^{\dim X - p + q}(X)$$

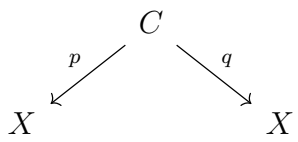
   which should be treated as an element in $H_{p-q}^{\mathrm{BM}}(X)$.

3. Poincare duality. For smooth X, we always have a fundamental class $[X] \in H_n^{\mathrm{BM}}(X)$. And cap with fundamental class gives me an isomorphism:

$$H^q(X) \simeq H_{\dim X - q}^{\mathrm{BM}}(X)$$

4. We can pushforward Borel-Moore homology class along a proper map. Geometrically, this is just integrating along the fibre.

Suppose we have a correspondence of $\mathbb{C}$-varieties:



with q *proper* and X *smooth*. Fix $\xi \in H_k^{\mathrm{BM}}(C)$, we can consider:

$$H^*(X) \xrightarrow{p^*} H^*(C) \xrightarrow{-\cap \xi} H_{k-*}^{\mathrm{BM}}(C) \xrightarrow{q_*} H_{k-*}^{\mathrm{BM}}(X) \simeq H^{2d-k+*}(X)$$

where $d = \dim_{\mathbb{C}} X$. In particular, if we take $k = 2d$, then we get a graded action on $H^*(X)$. For example, one might take ξ to be the fundamental class of a subvariety $Z \subset C$ of the same dimension as X. Anyway, we get a homomorphism:

$$H_{2d}^{\mathrm{BM}}(C) \to \mathrm{End}_{\mathrm{gr}}(H^*(X))$$

Remark. *If in addition C is an algebra / monoid object in the category of correspondences on X, that is, one equipped with a multiplication $C \times_X C \to C$ and a unit $X \hookrightarrow C$ with appropriate diagrams to commute. Then the map $H_{2d}^{\mathrm{BM}}(C) \to \mathrm{End}_{\mathrm{gr}}(H^*(X))$ is even a ring homomorphism.*

Let's make explicit what is the algebra struction on $H_{2d}^{\mathrm{BM}}(C)$ when C is an algebra object in the category of correspondences on X. Then the algebra structure is given by:

$$H_{2d}^{\mathrm{BM}}(C) \times H_{2d}^{\mathrm{BM}}(C) \xrightarrow{\boxtimes} H_{4d}^{\mathrm{BM}}(C \times C) \xrightarrow{\delta_X^!} H_{2d}^{\mathrm{BM}}(C \times_X C) \xrightarrow{m_*} H_{2d}^{\mathrm{BM}}(C)$$

where $\delta_X : X \hookrightarrow X \times X$ is the diagonal embedding and $\delta_X^!$ is the Gysin pullback.

6 Springer Action

Before developing the theory, let's quickly say what are going to be true.

1. If e is regular, then $H^*(\mathcal{B}_e)$ is the trivial representation.
2. For $G = \mathrm{GL}_n$, and $e = 0$. We actuallly have a W-action on the *space* $\mathcal{B}$ by the following. Let's fix a Hermitian inner product on $V = \mathbb{C}^n$, then a flag can be identified as an ordered set of orthogonal lines (of size n). Then $W = S_n$ simply permutes these factors.
3. More generally, we know $G/T \to G/B$ is an affine bundle. Then W clearly acts on $H^*(G/T) \simeq H^*(G/B)$. Note that a more canonical way to right G/T is:

   $$G/T \simeq \mathcal{BT} = \{(T, B) \mid T \subset B\}$$

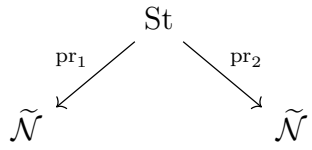
   the moduli space of pairs of maximal torus inside a Borel.
4. In fact $H^*(\mathcal{B})$ is the regular representaion of W, with the top degree giving the sign representaiton.
5. $H^2(\mathcal{B}) \simeq \mathfrak{t}^*$ is the dual of the abstract Cartan, on which W acts by reflections.
6. The W-action preserves the ring structure on cohomology.
7. $H^0(\mathcal{B})$ is the trivial representation. This is clear by connectedness of $\mathcal{B}$.

Example. *When $G = \mathrm{SL}_2$, $\mathcal{B} \simeq \mathbb{P}^1(\mathbb{C})$. Then Hopf fibration tells me:*

$$\mathbb{P}^1(\mathbb{C}) \simeq S^3/S^1 \simeq \mathrm{SU}(2)/T$$

Then $W \simeq N_{\mathrm{SU}(2)}(T)$ acts on $\mathbb{P}^1(\mathbb{C})$ via right translation. In particular $s \in W$ acts by antipodal map. In particular, it acts as -1 on $H^2(\mathbb{P}^1)$.

Let's use the general machinery of Borel-Moore homology in the following correspondence:



The algebra structure on St is given by:

$$\mathsf{St} \times_{\mathcal{N}} \mathsf{St} = \widetilde{\mathcal{N}} \times_{\widetilde{\mathcal{N}}} \widetilde{\mathcal{N}} \times_{\mathcal{N}} \widetilde{\mathcal{N}} \xrightarrow{\mathrm{pr}_{13}} \mathsf{St}$$

Therefore this gives me an algebra action:

$$\mathsf{H}^{\mathrm{BM}}_{2d}(\mathsf{St}) \to \mathrm{End}_{\mathrm{gr}}(H^*(\widetilde{\mathcal{N}})) \simeq \mathrm{End}_{\mathrm{gr}}(H^*(\mathcal{B}))$$

It is clear from geometry that $H^{\mathrm{BM}}_{2d}(\mathsf{St}) \simeq \mathbb{Q}[W]$ as vector spaces. However, the naive map:

$$w \mapsto [\mathsf{St}(w)]$$

is not an algebra morphism.

Example. *Consider $G = \mathrm{SL}_2$, $\mathcal{B} = \mathbb{P}^1$ and $\widetilde{\mathcal{N}} = T^*\mathbb{P}^1$. It is clear that $\mathsf{St}(1) \simeq \Delta\widetilde{\mathcal{N}}$, and*

$$\mathsf{St}(s) = \overline{\{(e, B, B') \mid B \neq B', e \in B \cap B'\}} \simeq \mathbb{P}^1 \times \mathbb{P}^1$$

Since if e is regular, then it lies in a unique Borel. This implies that $e = 0$ before taking closure. In this case, we have no restriction on choosing B, B'. It is true that $[\mathsf{St}(1)]$ corresponds to the unit, but one may compute:

$$[\mathsf{St}(s)] \cdot [\mathsf{St}(s)] = -2[\mathsf{St}(s)]$$

So the naive map is not an algebra morphism.

This suggests that we should try to find a different space in order to get the Springer action. Indeed, we should replace $\widetilde{\mathcal{N}}$ by the *Grothendieck alternation*.

Definition 6.1. The Grothendieck alternation $\widetilde{\mathfrak{g}}$ is given by the space:

$$\{(x, B) \in \mathfrak{g} \times \mathcal{B} \mid x \in \mathrm{Lie}(B)\}$$

This fits into the *almost* Cartesian diagram (up to nilpotent thickening)"

$$\begin{array}{ccc} \widetilde{\mathcal{N}} & \hookrightarrow & \widetilde{\mathfrak{g}} \\ \downarrow & & \downarrow \\ \mathcal{N} & \hookrightarrow & \mathfrak{g} \end{array}$$

Now, instead of just considering fibres over nilpotent elements, we consider the action on general fibres $\mathcal{B}_x$ for all $x \in \mathfrak{g}$.

Example. *Suppose $x \in \mathfrak{g}^{\mathrm{rs}}$, then $\mathcal{B}_x$ is just a W-torsor. Then the action of W on $H^*(\mathcal{B}_x)$ is clear. For type A, the W-torsor is explicit. Suppose we're considering:*

$$x = \mathrm{diag}(\lambda_1, \cdots, \lambda_n) \in \mathfrak{gl}_n$$

Then

$$\mathcal{B}_x = \{F_\bullet \mid xF_i \subset F_i\}$$

consists of flags $F_\bullet$ such that each F_i is a direct sum of eigenlines of x. This set is then in bijection with the total orderings of eigenvalues of x. This makes $\mathcal{B}_x$ an obvious S_n-torsor.

Now let's set:

$$\mathsf{St}_{\mathfrak{g}} = \widetilde{\mathfrak{g}} \times_{\mathfrak{g}} \widetilde{\mathfrak{g}}$$

Then we have the action morphism:

$$H^{\mathrm{BM}}_{2D}(\mathsf{St}_{\mathfrak{g}}) \to \mathrm{End}_{\mathrm{gr}}(H^*(\widetilde{\mathfrak{g}})) \simeq \mathrm{End}_{\mathrm{gr}}(H^*(\mathcal{B}))$$

where $D = \dim \mathfrak{g}$. Note that $\widetilde{\mathfrak{g}} \to \mathcal{B}$ is again a vector bundle (of fibre $\mathfrak{b}$), hence has the same cohomology groups. The geometry of $\mathsf{St}_{\mathfrak{g}}$ can be studied via studying its "easy" pieces. Namely:

$$\mathsf{St}_{\mathfrak{g}^{\mathrm{rs}}} = \widetilde{\mathfrak{g}}^{\mathrm{rs}} \times_{\mathfrak{g}} \widetilde{\mathfrak{g}^{\mathrm{rs}}} = \coprod_{w \in W} \Gamma(w|_{\widetilde{\mathfrak{g}^{\mathrm{rs}}}})$$

where $\Gamma(w|_{\widetilde{\mathfrak{g}^{\mathrm{rs}}}})$ is the graph of the action w on $\widetilde{\mathfrak{g}}^{\mathrm{rs}}$. We then denote $\mathsf{St}_{\mathfrak{g}}(w)$ the closure of $\Gamma(w|_{\widetilde{\mathfrak{g}}^{\mathrm{rs}}})$ in $\mathsf{St}_{\mathfrak{g}}$.

Theorem 6.2. *The irreducible components $\{\mathsf{St}_{\mathfrak{g}}(w)\}_{w \in W}$ exhausts all of the top dimensional irreducible components of $\mathsf{St}_{\mathfrak{g}}$.*

Remark. *It is actually true that $\mathsf{St}_{\mathfrak{g}}$ has pure dimension. But we won't need this fact.*

Proof. The content is a particular stratification of $\mathfrak{g}$. By Jordan decomposition, every element $x \in \mathfrak{g}$ can be written as:

$$x = x_s + x_n$$

For x_s, we should stratify according to "regularity", or in other word $H = C_G(x_s)$. Once we fix x_s, we can further conjugate x_n by $H = C_G(x_s)$. In other words, we should stratify $\mathfrak{g}$ according to discrete data $(H, \mathcal{O})$:

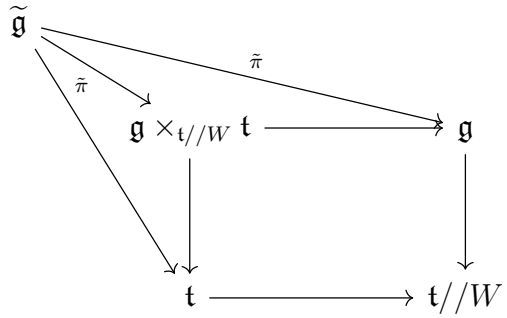
$$\mathfrak{g} = \coprod_{S=(H,\mathcal{O})/G} \mathfrak{g}_S$$

One now check that for any $x \in \mathfrak{g}_S$, we have:

$$\dim \mathcal{B}_x \leq \frac{1}{2}\mathrm{codim}_{\mathfrak{g}}\mathfrak{g}_S$$

with equality holding only when x is regular semisimple. This then implies that $\mathsf{St}_{\mathfrak{g}}(w)$ exhausts the top dimensional components of $\mathsf{St}_{\mathfrak{g}}$. □

Another important observation is that there exists a canonical map $\widetilde{\mathfrak{g}} \to \mathfrak{t}$ given by $(x, B) \to \overline{x}$. Here $\mathfrak{t}$ is the abstract Cartan. Therefore we have:



In fact, the map $\widetilde{\pi}$ is a resolution of singularity.

Example. *Consider $G = \mathrm{GL}_n$. Then $\widetilde{\mathfrak{g}} \to \mathfrak{t}$ is given by:*

$$(x, F_\bullet) \mapsto (\lambda_1, \cdots, \lambda_n)$$

where λ_i is the scalar that x acts by on F_i/F_{i-1}. The set $\mathfrak{g} \times_{\mathfrak{t}//W} \mathfrak{t}$ is given by a pair of element in $\mathfrak{g}$ and an ordering of its eigenvalues.

Example. *Let $G = \mathrm{SL}_2$, then:*

$$\mathfrak{g} \times_{\mathfrak{t}//W} \mathfrak{t} = \left\{ \left(\begin{pmatrix} a & b \\ c & -a \end{pmatrix}, \lambda \right) \mid \lambda^2 - a^2 - bc = 0 \right\}$$

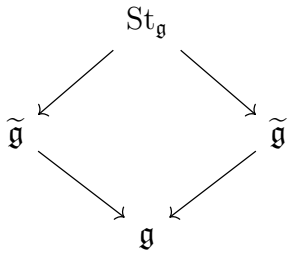
It has singularity at origin. If we blow up at the origin we get the special fibre $\mathbb{P}^1 \times \mathbb{P}^1$. However, for $\widetilde{\pi} : \widetilde{\mathfrak{g}} \to \mathfrak{g} \times_{\mathfrak{t}//W} \mathfrak{t}$, we have $\widetilde{\pi}^{-1}(0) \simeq \mathbb{P}^1$. This makes $\widetilde{\mathfrak{g}}$ a **small resolution**.

Now the restriction map determines an isomorphism:

$$H^{\mathrm{BM}}_{2D}(\mathsf{St}_{\mathfrak{g}}) \to H^{\mathrm{BM}}_{2D}(\mathsf{St}_{\mathfrak{g}^{\mathrm{rs}}}) \simeq \mathbb{C}[W]$$

where the last isomorphism comes from a W-torsor structure on $\mathsf{St}_{\mathfrak{g}^{\mathrm{rs}}} \to \mathfrak{g}^{\mathrm{rs}}$. This gives the Springer action of $\mathsf{St}_{\mathfrak{g}}$ on $H^{\mathrm{BM}}_*(\mathcal{B})$

Now let's construct the Springer action on $\mathcal{B}_e$ for nilpotent e. The idea is to consider the restriction of the diagram:



to the slowdoy slice.

Definition 6.3. Let (e, f, h) be a $\mathfrak{sl}_2$-triple. Then $S_e = e + \mathfrak{g}^f$ is called the *Slodowy slice* through e.

Note that S_e is a transversal slice. Indeed, note that:

$$T_e\mathcal{O}_e = \text{im } (\mathfrak{g} \xrightarrow{[e,-]} \mathfrak{g}), T_eS_e = \mathfrak{g}^f$$

By $\mathfrak{sl}_2$-theory we know $T_e\mathcal{O}_e \oplus T_eS_e = \mathfrak{g}$. Rephrasely differently, we can say $S_e \to [\mathfrak{g}/G]$ is a smooth map. This tells us that $\widetilde{S}_e$ is smooth fitting into:

$$\begin{array}{ccc} \widetilde{S}_e & \longrightarrow & [\widetilde{\mathfrak{g}}/G] \\ \downarrow & & \downarrow \\ S_e & \longrightarrow & [\mathfrak{g}/G] \end{array}$$

One can now similarly define:

$$\mathsf{St}_{S_e} = \widetilde{S}_e \times_{S_e} \widetilde{S}_e$$

whose irreducible components are closures of $\Gamma(w \mid_{\mathsf{St}_{S_e} \cap \mathfrak{g}^{\mathrm{rs}}})$. Therefore we have a natural map:

$$H^{\mathrm{BM}}_{2D}(\mathsf{St}_{\mathfrak{g}}) \simeq H^{\mathrm{BM}}_{2D_e}(\mathsf{St}_{S_e}) \to \mathrm{End}_{\mathrm{gr}}(H^*(\widetilde{S}_e)) \simeq \mathrm{End}_{\mathrm{gr}}(H^*(\mathcal{B}_e))$$

The last isomorphism comes from the fact we have a $\mathbb{G}_m$-action that contracts $\widetilde{S}_e$ to $\mathcal{B}_e$ (and S_e to e).

In fact the action on S_e is given by formula:

$$t \cdot (e + x) = t^2 \mathrm{Ad}(h(t^{-1}))(e + x)$$

where we treat $h : \mathbb{G}_m \to G_{\mathrm{ad}}$ and thus e has weigth 2, f has weight -2, $\mathfrak{g}^f$ has weight ≤ 0. Then:

$$\lim_{t \to 0} t \cdot (e + x) = e$$

The lifted action on $\widetilde{S}_e$ is then given by:

$$t \cdot (e + x, B) = (t^2 \mathrm{Ad}(h(t^{-1}))(e + x), \mathrm{Ad}(h(t^{-1}))B)$$

Another consequence for the contraction is that we can prove the purity of cohomology of Springer fibres.

Theorem 6.4 (Springer)**.** *$H^*(\mathcal{B}_e)$ is pure in the sense of mixed Hodge structure (over $\mathbb{C}$) or weights of Frobenius (over $\mathbb{F}_q$).*

The idea of the proof is using the fact that $\widetilde{S}_e$ is smooth and thus $H^i(\widetilde{S}_e)$ has weight $\geq i$. But $\mathcal{B}_e$ is proper so that $H^i(\mathcal{B}_e)$ has weight $\leq i$.

Theorem 6.5 (DeCocini-Lusztig-Procesi)**.** *We have $H^{\mathrm{odd}}(\mathcal{B}_e) = 0$ and $H^*(\mathcal{B}_e)$ is spanned by algebraic cycles.*

7 Perverse Sheaves

This section is devoted to have a quick review of the theory of perverse sheaves. The way we will use the theory is the following.

Let $D(\mathcal{N})$ be the *bounded derived category of constructed complexes on $\mathcal{N}/\mathbb{C}$ with coefficients in $\mathbb{Q}$*. Now if $\nu : \widetilde{\mathcal{N}} \to \mathcal{N}$ is the Springer resolution, we consider:

$$\nu_* \underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}} \in D(\mathcal{N})$$

Basic base change theorem tells me:

$$(\nu_* \underline{\mathbb{Q}})_e \simeq H^\bullet(\mathcal{B}_e, \mathbb{Q})$$

We will eventually show:

$$\mathrm{End}(\nu_* \underline{\mathbb{Q}}) \simeq \mathbb{Q}[\mathbb{W}]$$

and therefore $\mathbb{Q}[\mathbb{W}]$ acts on $H^\bullet(\mathcal{B}_e, \mathbb{Q})$.

Now let X be a topological space, and we fix a Whitney stratification $\mathcal{S}$ on X. Denote $D_{\mathcal{S}}(X)$ be the bounded derived category of complexes of $\mathbb{Q}$-sheaves, constructible with respect to $\mathcal{S}$. Let's describe six functor formalism explicitly in specific cases.

Let $\iota_\alpha : X_\alpha \hookrightarrow X$ be the natural inclusion. We have:

$$\iota_\alpha^* \mathcal{F} = \mathcal{F}|_{X_\alpha}; \quad \iota_\alpha^! \mathcal{F} \in D(X_\alpha, \mathbb{Q})$$

Example. *If $X_\alpha = \{x\}$ is singleton, then:*

$$\iota_\alpha^* \mathcal{F} = \mathcal{F}_x$$

is the **stalk**. *Note this is an exact functor. On the other hand:*

$$\iota_\alpha^! \mathcal{F} = R\Gamma_{\{x\}}(U, \mathcal{F})$$

is the **costalk**. *This fits into distinguished triangle*

$$R\Gamma_{\{x\}}(U, \mathcal{F}) \to R\Gamma(U, \mathcal{F}) \to R\Gamma(U - \{x\}, \mathcal{F}) \xrightarrow{[1]}$$

This generalizes to the general distinguished triangle:

$$Z \xrightarrow{i} X \xleftarrow{j} U$$

$$i_* i^! \mathcal{F} \to \mathcal{F} \to j_* j^* \mathcal{F} \to$$

Or dually:

$$j_! j^* \mathcal{F} \to \mathcal{F} \to i_* i^* \mathcal{F}$$

In practice, the way to compute $\iota_\alpha^! \mathcal{F}$ is by taking a transversal slice S through x and:

$$(\iota_\alpha^! \mathcal{F})_x \simeq R\Gamma_{\{x\}}(S, \mathcal{F}|_S)$$

Now let's enter the realm of algebraic geometry. We assume X to be a *complex* algebraic variety, and $\mathcal{S}$ is a stratification such that each strata X_α is smooth connected of complex dimension d_α. In the remaining of this section all dimensions are *complex dimensions*.

Definition 7.1. Consider:

$$^pD^{\leq 0}_{\mathcal{S}} \coloneqq \{\mathcal{F} \mid \iota_\alpha^*\mathcal{F} \text{ concentrates in degrees} \leq -d_\alpha\}$$

$$^pD^{\geq 0}_{\mathcal{S}} \coloneqq \{\mathcal{F} \mid \iota_\alpha^!\mathcal{F} \text{ concentrates in degrees} \geq -d_\alpha\} = \mathbb{D}(^pD^{\leq 0}_{\mathcal{S}})$$

Then the perverse sheaves are defined to be:

$$\mathrm{Perv}_{\mathcal{S}}(X) = {}^pD^{\leq 0}_{\mathcal{S}} \cap {}^pD^{\geq 0}_{\mathcal{S}}$$

These two truncations also define a t-structure on the bounded derived category of constructible sheaves, called the **perverse t-structure**. We can therefore also define **perverse cohomology**.

The next step is to construct perverse sheaves. This can be done via intermediate extensions. Let $\iota_\alpha : X_\alpha \to X$. We wish to define:

$$\iota_{\alpha,!*} : \mathrm{Perv}(X_\alpha) \to \mathrm{Perv}(X)$$

Let's explain Deligne's construction, in the case of 3-step stratificaiton: $X = X_\alpha \coprod X_\beta \coprod X_\gamma$ where X_α is closed, X_γ is open, X_β is somewhere in middle.

1. First we produce a perverse sheaf on $X_\alpha \coprod X_\beta$. Let $\iota_\alpha^{\alpha,\beta} : X_\alpha \hookrightarrow X_\alpha \coprod X_\beta$, and we take:
$$\tau^{\beta}_{\leq -d_\beta - 1}(\iota_\alpha^{\alpha,\beta})_*\mathcal{L}[d_\alpha]$$
One should check this is a perverse sheaf on $X_\alpha \coprod X_\beta$. Note we're truncating one step better than the perversity condition requires.

2. One then repeat this process. Let $\iota_{\alpha,\beta}^{\alpha,\beta,\gamma} : X_\alpha \coprod X_\beta \hookrightarrow X$, and we take:
$$\tau^{\gamma}_{\leq -d_\gamma - 1}(\iota_{\alpha,\beta}^{\alpha,\beta,\gamma})_*(\tau^{\beta}_{\leq -d_\beta - 1}(\iota_\alpha^{\alpha,\beta})_*\mathcal{L}[d_\alpha])$$
This will be our final intermediate extension:
$$\iota_{\alpha,!*}\mathcal{L}[d_\alpha]$$

The explicit construction is not useful. Instead, we will need:

Proposition 7.2. *The intermediate extension $\mathcal{F} = (\iota_\alpha)_{!*}\mathcal{L}[d_\alpha]$ are characterized by properties:*

1. *$\mathcal{F}$ is supported on $\overline{X_\alpha}$.*
2. *$\iota_\beta^*\mathcal{F}$ lies in degrees $\leq -d_\beta - 1$ for all $X_\beta \subset \overline{X_\alpha}$, $\beta \neq \alpha$*
3. *$\iota_\beta^!\mathcal{F}$ lies in degree $\geq -d_\beta + 1$ for all $X_\beta \subset \overline{X_\alpha}$, $\beta \neq \alpha$.*
4. *$\mathcal{F}|_{X_\alpha} = \mathcal{L}[d_\alpha]$*

Theorem 7.3. *We have $K_0(\mathrm{Perv}_{\mathcal{S}}(X))$ as an abelian group is freely generated by:*

$$(\iota_\alpha)_{!*}\mathcal{L}[d_\alpha]$$

for $\mathcal{L} \in \mathrm{LS}(X_\alpha)$ irreducible.

Now suppose G acts on X, inducing a stratification by orbits. Then the G-equivariant perverse sheaves are even easier to describe. For an orbit $e \in \mathcal{O}$, we have $\mathcal{O} \simeq G/Z_G(e)$. Therefore a G-equivariant local system on $\mathcal{O}$ is nothing but a continuous representation of $Z_G(e)$, or in other word a representation of $A_e = \pi_0(Z_G(e))$.

Remark. *If G is simply connected, then so is $G/Z_G(e)^\circ$. So that:*

$$G/Z_G(e)^\circ \to \mathcal{O}$$

is an universal cover. Therefore we have $\pi_1(\mathcal{O}) \simeq \pi_0(Z_G(e)) = A_e$.

Example. *Let $\mathcal{N}$ be the nilpotent orbit of GL_2 or PGL_2. Then:*

$$\mathrm{Perv}_{\mathrm{GL}_2}(\mathcal{N}) \neq \mathrm{Perv}_{\mathrm{PGL}_2}(\mathcal{N})$$

Note for the former we have $(\mathcal{O}_{\mathrm{reg}}, \mathsf{sgn})$.

Finally let's discuss the general condition to ensure pushforward of shifted local system still being a perverse sheaf.

Definition 7.4. Let X, Y be irreducible varieties. A dominant map $f : X \to Y$ is called:

1. **small** if for all $i > 0$, the closed subset:

$$\{y \in Y \mid \dim f^{-1}(y) \geq i\}$$

   has codimension $\geq 2i + 1$ in Y.

2. **semismall** if for all $i \geq 0$, the above closed subset has codimension $\geq 2i$.

Remark. *An equivalent characterization of semismallness is the condition that $\dim(X \times_Y X) = \dim X$. The characterization for small map is more complicated. It's equivalent to say that if V is some (or any) open subset of Y such that*

$$f : U \coloneqq f^{-1}(V) \to V$$

is quasi-finite, then f is small iff

$$\dim(X \setminus U) \times_{Y \setminus V} (X \setminus U) < \dim X$$

Example. *$\pi : \widetilde{\mathfrak{g}} \to \mathfrak{g}$ is small; $\nu : \widetilde{\mathcal{N}} \to \mathcal{N}$ is semismall.*

Theorem 7.5. *Let $f : X \to Y$ be a proper dominant morphism between irreducible varieties.*

1. If f is semismall, then pushforward of a perverse sheaf is again perverse.

2. If in addition f is small, then pushforward of IC sheaves are IC sheaves.

There is corresponding version for equivariant perverse sheaves.

Finally let's discuss decomposition theorem.

Theorem 7.6 (Decomposition theorem)**.** *Let $f : X \to Y$ be projective where X is smooth of dimension n. Then:*

$$Rf_*\underline{\mathbb{Q}}$$

is a direct sum of shifted of simple perverse sheaf.

When f is semismall, then we know there need not be shifts. The content of the theorem is the complete reducibility.

Note this theorem depends on the algebraic nature of the map. In particular, it is not necessarily true in analytic setting.

8 Springer Correspondence

Let $d = \dim \mathfrak{g}$. Consider Grothendieck alteration:

$$\pi : \widetilde{\mathfrak{g}} \to \mathfrak{g}$$

This is a small map restricting to a $\mathbb{W}$-torsor on $\mathfrak{g}^{\mathrm{rs}}$. Therefore if we set:

$$\mathcal{L} = (\pi|_{\widetilde{\mathfrak{g}}^{\mathrm{rs}}})_*\underline{\mathbb{Q}} \in \mathrm{LS}(\mathfrak{g}^{\mathrm{rs}})$$

Then we know that:

$$\pi_*\underline{\mathbb{Q}}[d] = j_{!*}\mathcal{L}[d]$$

Fact: For any locally closed embedding h, we have:

$$\mathrm{Hom}(h_{!*}\mathcal{L}_1, h_{!*}\mathcal{L}_2) \simeq \mathrm{Hom}(\mathcal{L}_1, \mathcal{L}_2)$$

Using this, we know:

$$\mathrm{End}(\pi_*\underline{\mathbb{Q}}[d]) \simeq \mathrm{End}(\mathcal{L})$$

But recall the category of local systems on a space X is equivalent to the category of representations of $\pi_1(X, x)$ by taking the fibre at $x \in X$. So that:

$$\mathrm{End}(\mathcal{L}) \simeq \mathrm{End}_{\pi_1(\mathfrak{g}^{\mathrm{rs}}, x)}(\mathbb{Q}^{|\mathbb{W}|}) \simeq \mathrm{End}_{\mathbb{W}}(\mathbb{Q}^{|\mathbb{W}|}) \simeq \mathbb{Q}[\mathbb{W}]$$

Moreover, we can decompose $\mathcal{L}$ into simple local systems:

$$\mathcal{L} = \bigoplus_{\rho} \mathcal{L}_\rho \otimes \rho$$

where $\mathcal{L}_\rho$ is the simple local system corresponding to $\mathbb{W}$-module ρ and the right copy $\rho \simeq \rho^*$ is the multiplicity space (which also admits a $\mathbb{W}$-action). Based on this, we define:

$$\mathcal{E}_\rho = j_{!*}(\mathcal{L}_\rho[d]), \quad j : \mathfrak{g}^{\mathrm{rs}} \hookrightarrow \mathfrak{g}$$

and therefore:

$$\pi_*\underline{\mathbb{Q}}[d] = \bigoplus_{\rho\in\mathrm{Irr}(\mathbb{W})} \rho\otimes\mathcal{E}_\rho$$

Now π restricts to Springer resolution $\nu : \widetilde{\mathcal{N}} \to \mathcal{N}$. Let $r = \mathrm{rk}\ \mathfrak{g}$, then $\dim\mathcal{N} = d - r$. Then:

$$\nu_*\underline{\mathbb{Q}}[d-r] = \pi_*\underline{\mathbb{Q}}[d-r]|_{\mathcal{N}} = \bigoplus_{\rho\in\mathrm{Irr}(\mathbb{W})} \rho\otimes\mathcal{E}_\rho[-r]|_{\mathcal{N}}$$

Note that we know $\mathcal{F}_\rho \coloneqq \mathcal{E}_\rho[-r]|_{\mathcal{N}} \in \mathrm{Perv}_G(\mathcal{N})$ is a perverse sheaf, since by semismallness of ν we know $\nu_*\underline{\mathbb{Q}}[d-r]$ is perverse.

Theorem 8.1. *The perverse sheaf $\mathcal{F}_\rho$ is simple.*

By decomposition theorem, we know $\mathcal{F}_\rho$ is a direct sum of simple perverse sheaves. To show it's actually simple, we expect to show the restriction map:

$$\mathbb{Q}[\mathbb{W}] \simeq \mathrm{End}(\pi_*\underline{\mathbb{Q}}) \to \mathrm{End}(\nu_*\underline{\mathbb{Q}})$$

is an isomorphism. First of all, we need:

Lemma 8.2. $\dim(\mathrm{End}(\nu_*\underline{\mathbb{Q}})) = |\mathbb{W}|$.

Proof. We use proper base change. Consider the diagram:

$$\begin{array}{ccc} \mathrm{St} & \xrightarrow{\ p\ } & \widetilde{\mathcal{N}} \\ \downarrow{\scriptstyle q} & & \downarrow{\scriptstyle \nu} \\ \widetilde{\mathcal{N}} & \xrightarrow{\ \nu\ } & \mathcal{N} \end{array}$$

Since ν is proper, we compute:

$$\mathrm{End}(\nu_*\underline{\mathbb{Q}}) \simeq \mathrm{Hom}(\nu_!\underline{\mathbb{Q}}, \nu_*\underline{\mathbb{Q}}) \simeq \mathrm{Hom}(\underline{\mathbb{Q}}, \nu^!\nu_*\underline{\mathbb{Q}}) \simeq \mathrm{Hom}(\underline{\mathbb{Q}}, q_*p^!\underline{\mathbb{Q}}) \simeq \mathrm{Hom}(\underline{\mathbb{Q}}, p^!\underline{\mathbb{Q}})$$

Using that $\widetilde{\mathcal{N}}$ is smooth, we know $\omega_{\widetilde{\mathcal{N}}} = \underline{\mathbb{Q}}[2\dim\mathcal{N}]$ and therefore:

$$p^!\underline{\mathbb{Q}} = \omega_{\mathsf{St}}[-2\dim\mathcal{N}]$$

Therefore:

$$\mathrm{End}(\nu_*\underline{\mathbb{Q}}) = \mathrm{RHom}^{-2\dim\mathcal{N}}(\underline{\mathbb{Q}}, \omega_{\mathsf{St}}) = H^{-\mathrm{top}}(\mathsf{St}, \omega_{\mathsf{St}}) = H^{\mathrm{BM}}_{\mathrm{top}}(\mathsf{St})$$

We are therefore done because we understood the geometry of St before. □

In particular, we construct a $H^{\mathrm{BM}}_{\mathrm{top}}(\mathsf{St})$ action on $\nu_*\underline{\mathbb{Q}}$, which is the localized version of its action on $H^*(\widetilde{\mathcal{N}})$. This is because:

$$H^k(\mathsf{St}, \nu_*\underline{\mathbb{Q}}) = \mathrm{Hom}(\underline{\mathbb{Q}}, \nu_*\underline{\mathbb{Q}}[k]) = \mathrm{Hom}_{\widetilde{\mathcal{N}}}(\underline{\mathbb{Q}}, \underline{\mathbb{Q}}[j]) = H^k(\widetilde{\mathcal{N}})$$

Note that the same computation tells me I have the following diagram:

$$\begin{array}{ccc} \mathrm{End}(\pi_*\underline{\mathbb{Q}}) & \xrightarrow{\sim} & H^{\mathrm{BM}}_{\mathrm{top}}(\mathrm{St}_{\mathfrak{g}}) \\ \downarrow \sim & & \downarrow \\ \mathrm{End}(\nu_*\underline{\mathbb{Q}}) & \xrightarrow[\sim]{} & H^{\mathrm{BM}}_{\mathrm{top}}(\mathrm{St}) \end{array}$$

However, the natural map:

$$H^{\mathrm{BM}}_{\mathrm{top}}(\mathsf{St}_{\mathfrak{g}}) \to H^{\mathrm{BM}}_{\mathrm{top}}(\mathsf{St})$$

will **not** map the natural basis $[\Gamma_w]$ to $[\mathsf{St}_w]$. In fact, the transition matrix under these two bases are called *microstalks*.

We also need:

Lemma 8.3. $\mathcal{F}_\rho \neq 0$

Proof. We will show that $(\mathcal{F}_\rho)_0 \neq 0$. This can be seen via:

$$H^*(\mathcal{B}) \simeq (\nu_*\underline{\mathbb{Q}})_0 = \bigoplus_\rho \rho \otimes (\mathcal{F}_\rho[r-d])_0$$

as $\mathbb{W}$-modules. But LHS is just regulr representation, this means that $\mathcal{F}_{\rho,0} \neq 0$. We are done. □

Now an easy dimension counting tells us that each $\mathcal{F}_\rho$ are distinct and they're all simple. As a consequence, we have an injection:

$$\mathrm{Irr}(\mathbb{W}) \hookrightarrow \mathrm{Irr}(\mathrm{Perv}_G(\mathcal{N})) = \{(\mathcal{O}, \chi) \mid \mathcal{O} \in G \setminus \mathcal{N}, \chi \in \mathrm{Irr}(A_\mathcal{O})\}$$

This map is not necessarily surjective. To state the full Springer correspondence, we need to determine its (partial) inverse.

Proposition 8.4. *Given* $(\mathcal{O}, \chi) \in \mathrm{Irr}(\mathrm{Perv}_G(\mathcal{N}))$*, we consider:*

$$E_{\mathcal{O},\chi} \coloneqq \mathrm{Hom}_{A_e}(\chi, H^{\mathrm{top}}(\mathcal{B}_e))$$

Then either $E_{\mathcal{O},\chi}$ *is either an irrep of* $\mathbb{W}$ *or equals* 0*.*

Proof. The idea is again to write cohomology in a sheafy manner. Note that:

$$H^*(\mathcal{B}_e) = (\nu_*\underline{\mathbb{Q}})_e = \bigoplus_\rho \rho \otimes (\mathcal{E}_\rho[-d])_e = \bigoplus_\rho \rho \otimes (\mathcal{F}_\rho)_e[-(d-r)]$$

Since F_ρ is a simple perverse sheaf, we know there exists $(\mathcal{O}, \chi)$ such that $F_\rho = \mathrm{IC}(\mathcal{O}, \chi)$. It is clear that if $(\mathcal{F}_\rho)_e \neq 0$ then $\mathcal{O}_e \leq \mathcal{O}$ in the closure ordering. But now we're taking taking top degree cohomology (note that here the top degree is $2 \cdot \dim \mathcal{B}_e = \mathrm{codim}_\mathcal{N}(\mathcal{O}_e)$), so that:

$$H^{\mathrm{top}}(\mathcal{B}_e) = \bigoplus_{\rho:\mathcal{O}_\rho \geq \mathcal{O}_e} \rho \otimes H^{\mathrm{codim}_\mathcal{N}(\mathcal{O}_e)-(d-r)}(\mathrm{IC}(\mathcal{O}, \chi)_e) = \bigoplus_{\rho:\mathcal{O}_\rho \geq \mathcal{O}_e} \rho \otimes H^{-\dim \mathcal{O}_e}(\mathrm{IC}(\mathcal{O}, \chi)_e)$$

For $\mathcal{O} > \mathcal{O}_e$, the nature of IC sheaves tells me that:

$$\mathrm{IC}(\mathcal{O}, \chi)|_{\mathcal{O}_e}$$

concentrates in degree $\leq -\dim \mathcal{O}_e - 1$, making the cohomology group vanishes. Therefore we must have $\mathcal{O} = \mathcal{O}_e$ and so that $\mathrm{IC}(\mathcal{O}, \chi)|_{\mathcal{O}} = \chi[\dim \mathcal{O}]$. Hence we have:

$$H^{\mathrm{top}}(\mathcal{B}_e) = \bigoplus_{\rho : \mathcal{O}_\rho = \mathcal{O}_e} \rho \otimes \chi_\rho$$

Therefore if $\chi = \chi_\rho$ for some ρ, then:

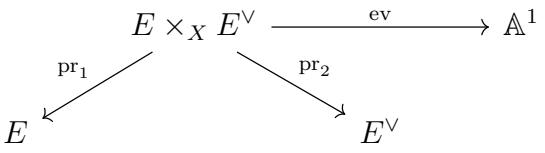
$$E_{\mathcal{O},\chi} = \mathrm{Hom}_{A_e}(\chi_\rho, H^{\mathrm{top}})(\mathcal{B}_e) = \rho$$

as required. □

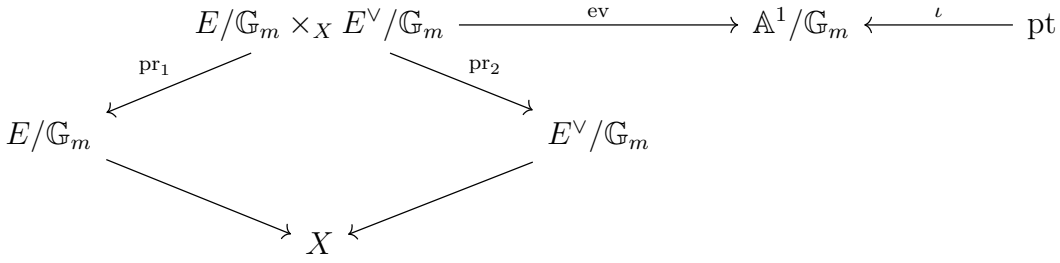
9 Fourier Transform on Perverse Sheaves

Recall the way we prove $\mathcal{F}_\rho$ is simple is via decomposition theorem. Crucially, we translate the problem to a more classical setting involving Borel-Moore homology of different types of Steinberg varieties. In order to provide a proof in a purely sheaf theoretical manner, we need to introduce **Fourier transform on perverse sheaves**.

Let's start from the general setting of Fourier transform. Let X be a variety with vector bundle E. Let $E^\vee$ be the dual vector bundle. Let's consider the following diagram:



After equipped with compatible $\mathbb{G}_m$-actions:



where ι is the embedding of the nonstacky point. Now let's defined the **integral kernel** to be:

$$\mathrm{ev}^* \iota_* \underline{\mathbb{Q}}_{\mathrm{pt}}$$

And we define the Fourier transform to be:

$$\mathrm{FT} : D_{\mathbb{G}_m}(E) \to D_{\mathbb{G}_m}(E^\vee)$$

$$\mathrm{FT}(\mathcal{F}) = \mathrm{pr}_{2!}(\mathrm{pr}_1^* \mathcal{F} \otimes^{\mathbb{L}} \mathrm{ev}^* \iota_* \underline{\mathbb{Q}}_{\mathrm{pt}})[r-1]$$

where r is the rank of the vector bundle. Let's list some basic facts about Fourier transform:

1. Fourier transform is an involution up to some Tate twist. In particular, it is an equivalence of categories.

2. Fourier transform is t-exact with respect to the perverse t-structure. In particular, we have:
$$\mathrm{FT} : \mathrm{Perv}_{\mathbb{G}_m}(E) \to \mathrm{Perv}_{\mathbb{G}_m}(E^\vee)$$

3. Fourier transform behaves well under base change. Consider the setting:
$$\begin{array}{ccccc} E_X & \longrightarrow & X & \longleftarrow & E_X^\vee \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle \phi} & & \downarrow{\scriptstyle f^\vee} \\ E & \longrightarrow & Y & \longleftarrow & E^\vee \end{array}$$
then we have:
$$\mathrm{FT}(f_*\mathcal{F}) = f_*^\vee \mathrm{FT}(\mathcal{F})$$
$$\mathrm{FT}(f_!\mathcal{F}) = f_!^\vee \mathrm{FT}(\mathcal{F})$$

4. Under sheaf-function correspondence we recover standard Fourier transform. A particular instance is the following. For $S \subset E$ a subbundle, with dual bunle $S^\perp \subset E^\vee$. Then:
$$\mathrm{FT}(\underline{\mathbb{Q}}_S[\dim S]) = \underline{\mathbb{Q}}_{S^\perp}[\dim S^\perp]$$
In particular,
$$\mathrm{FT}(\underline{\mathbb{Q}}_E[\dim E]) = \mathbb{Q}_X[\dim X]$$

Let's now consider the diagram:
$$\begin{array}{ccccccccc} \widetilde{\mathfrak{g}} & \hookrightarrow & \mathfrak{g}\times\mathcal{B} & \longrightarrow & \mathcal{B} & \longleftarrow & \mathfrak{g}^*\times\mathcal{B} & \hookleftarrow & \widetilde{\mathcal{N}} \\ & {\scriptstyle\pi}\searrow & \downarrow{\scriptstyle\tilde{\pi}} & & \downarrow & & \downarrow{\scriptstyle\tilde{\pi}^\vee} & & \downarrow{\scriptstyle\nu} \\ & & \mathfrak{g} & \longrightarrow & \mathrm{pt} & \longleftarrow & \mathfrak{g}^* & \hookleftarrow & \mathcal{N} \end{array}$$

Here the middle two squares fits into the setting of base change. Moreover, $\widetilde{\mathfrak{g}}^\perp = \widetilde{\mathcal{N}}$ are dual to each other. Therefore we see that:
$$\mathrm{FT}(\pi_*\underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}}}[d]) = \mathrm{FT}(\widetilde{\pi}_*\underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}}}[d]) = \widetilde{\pi}_*^\vee \mathrm{FT}(\underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}}}[d]) = \widetilde{\pi}_*^\vee \underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}}[d-r] = \nu_*\underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}}[d-r]$$

But recall:
$$\nu_*\underline{\mathbb{Q}}_{\widetilde{\mathcal{N}}}[d-r] = \bigoplus_{\rho\in\mathrm{Irr}(\mathbb{W})} \rho\otimes\mathcal{F}_\rho$$
$$\pi_*\underline{\mathbb{Q}}_{\widetilde{\mathfrak{g}}}[d] \simeq \bigoplus_{\rho\in\mathrm{Irr}(\mathbb{W})} \rho\otimes\mathcal{E}_\rho$$
with each $\mathcal{E}_\rho$ simple. Since Fourier transform is an equivalence, this forces each F_ρ to be simple by counting (provided we show that each $\mathcal{F}_\rho \neq 0$).

Example. *Let's take $\rho = \mathsf{triv}$, then $\mathcal{L}_{\mathsf{triv}} \in \mathrm{LS}(\mathfrak{g}^{\mathrm{rs}})$ is the trivial local system. Then $\mathcal{E}_\rho = j_{!*}(\mathcal{L}_{\mathsf{triv}}[d]) = \underline{\mathbb{Q}}_{\mathfrak{g}}[d]$. Now we know the restriction (after homological shift) of a simple perverse sheaf on $\mathfrak{g}$ is still perverse; in particular: $\underline{\mathbb{Q}}_{\mathcal{N}}[d-r]$ is simple perverse sheaf, which implies that $\mathcal{N}$ is rationally smooth.*

Example. *Now let's compute some Fourier transform.*

$$\mathrm{FT}(\mathcal{E}_{\mathsf{triv}}) = \mathrm{FT}(\underline{\mathbb{Q}}_{\mathfrak{g}}[d]) = \delta_0$$

It's a nontrivial computation shows that:

$$\mathcal{E}_{\mathsf{sgn}}|_{\mathcal{N}} = \delta_0$$

In general, we will have:

$$\mathrm{FT}(\mathcal{E}_\rho) = \mathcal{E}_{\rho\otimes\mathsf{sgn}}|_{\mathcal{N}}[?\,]$$

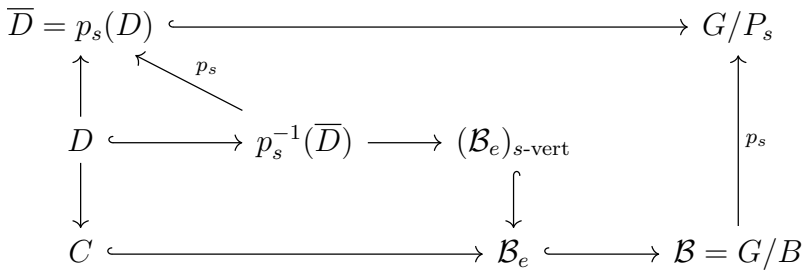
10 Hotta's Formula

Hotta's formula makes the computation for Springer representations explicit. Suppose we want to explicitly describe the action of $\mathbb{W}$ on $H^{\mathrm{top}}(\mathcal{B}_e) = H_{2d_e}(\mathcal{B}_e)$, here $d_e = \dim_{\mathbb{C}} \mathcal{B}_e$ is the dimension of the Springer fibre. We only need to describe the action of a simple reflection s.

Consider the map $p_s : G/B \to G/P_s$.

Definition 10.1. A component $C \subset \mathcal{B}_e$ is called *s-horizontal* if $p_s(C) \simeq C$; and otherwise we call C *s-vertical*.

If C is s-vertical, then $s \cdot [C] = -[C]$. If C is s-horizontal, then $s \cdot [C] = [C] +$ (something vertical). We need to determine what is "something vertical". Let's consider the following diagram:



Here $D = C \cap (\mathcal{B}_e)_{s\text{-vert}}$ is a divisor in $\mathcal{B}_e$. One can readily see that $p_s^{-1}(\overline{D})$ is the union of a some components of $(\mathcal{B}_e)_{s\text{-vert}}$. Therefore we have:

$$p_{s,*}[D] \in H_{2d_e-2}(\overline{D})$$

Now that using the fact that:

$$p_s : p_s^{-1}(\overline{D}) \to \overline{D}$$

is a $\mathbb{P}^1$-bundle, we can apply Gysin pullback:

$$p_s^!(p_{s,*}[D]) \in H_{2d_e}(p_s^{-1}(\overline{D})) \subset H_{2d_e}((\mathcal{B}_e)_{\mathrm{vert}})$$

This gives the vertical contribution. Geometrically, Gysin pullback is simply take preimage of a class.

Example. *Let's consider $G = \mathrm{SL}_3$, and e be subregular. Then $\mathcal{B}_e$ is union of two copies of $\mathbb{P}^1$. For any simple reflection s, one copy of $\mathbb{P}^1$ will be horizontal and the other copy will be vertical. Let the vertical one be C_1 and the horizontal one be C_2. Then we know:*

$$s \cdot [C_1] = -[C_1]$$

Moreover, $D = C_1 \cap C_2 = \mathrm{pt}$. From this it's evident that:

$$p_s^!(p_{s,*}[D]) = [C_1] \in H_2(C_1)$$

and therefore: $s \cdot [C_2] = [C_2] + [C_1]$.